

Day 11 (part 2): Static Routing

REVIEW:

- SWITCHES forward traffic WITHIN LAN's
- ROUTERS forward traffic BETWEEN LAN's

WAN (Wide Area Network)

- Network spread over a large area



R2 Connected & Local Routes

```
R2# conf t
R2(config)# interface g0/0
R2(config-if)# ip address 192.168.12.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# interface g0/1
R2(config-if)# ip address 192.168.24.2 255.255.255.0
R2(config-if)# no shutdown

R2# show ip route
!codes output omitted

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet0/0
L    192.168.12.2/32 is directly connected, GigabitEthernet0/0
C    192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.24.0/24 is directly connected, GigabitEthernet0/1
L    192.168.24.2/32 is directly connected, GigabitEthernet0/1
```

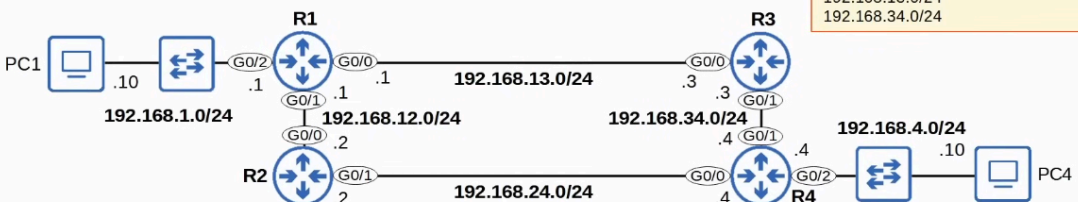
The following routes are automatically added to the routing table for each interface with an IP address configured:

- **C - Connected**
→ A route to the network the interface is connected to. (with the actual netmask configured on the interface)
- **L - Local**
→ A route to the actual IP address configured on the interface. (with a /32 netmask)

R2 knows how to reach its own IP addresses and destinations in its connected networks, but it doesn't know how to reach destinations in remote networks.

Knows:
192.168.12.0/24 (incl. 192.168.12.2/32)
192.168.24.0/24 (incl. 192.168.24.2/32)

Doesn't know:
192.168.1.0/24
192.168.4.0/24
192.168.13.0/24
192.168.34.0/24



Network diagram showing four routers (R1, R2, R3, R4) and two PCs (PC1, PC4). R1 is connected to R2 and R3. R2 is connected to R4. R3 is connected to R4. PC1 is connected to R1. PC4 is connected to R4. IP addresses and subnet masks are shown for each interface.

R3 Connected & Local Routes

```
R3# conf t
R3(config)# interface g0/0
R3(config-if)# ip address 192.168.13.3 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# interface g0/1
R3(config-if)# ip address 192.168.34.3 255.255.255.0
R3(config-if)# no shutdown
```

```
R3# show ip route
!codes output omitted
```

```
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/0
L    192.168.13.3/32 is directly connected, GigabitEthernet0/0
192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet0/1
L    192.168.34.3/32 is directly connected, GigabitEthernet0/1
```

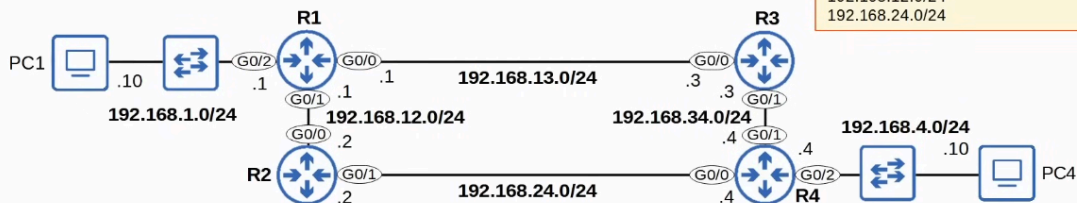
R3 knows how to reach its own IP addresses and destinations in its connected networks, but it doesn't know how to reach destinations in remote networks.

Knows:

192.168.13.0/24 (incl. 192.168.13.3/32)
192.168.34.0/24 (incl. 192.168.34.3/32)

Doesn't know:

192.168.1.0/24
192.168.4.0/24
192.168.12.0/24
192.168.24.0/24



R4 Connected & Local Routes

```
R4(config)# interface g0/0
R4(config-if)# ip address 192.168.24.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# interface g0/1
R4(config-if)# ip address 192.168.34.4 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# interface g0/2
R4(config-if)# ip address 192.168.4.4 255.255.255.0
R4(config-if)# no shutdown
```

```
R4# show ip route
!codes output omitted
```

```
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/2
L    192.168.4.4/32 is directly connected, GigabitEthernet0/2
192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.24.0/24 is directly connected, GigabitEthernet0/0
L    192.168.24.4/32 is directly connected, GigabitEthernet0/0
192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet0/1
L    192.168.34.4/32 is directly connected, GigabitEthernet0/1
```

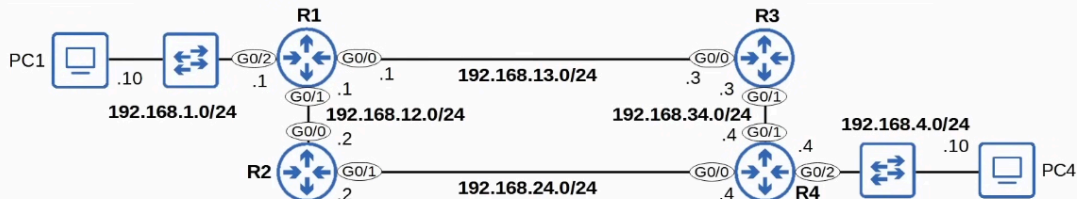
R4 knows how to reach its own IP addresses and destinations in its connected networks, but it doesn't know how to reach destinations in remote networks.

Knows:

192.168.4.0/24 (incl. 192.168.4.4/32)
192.168.24.0/24 (incl. 192.168.24.4/32)
192.168.34.0/24 (incl. 192.168.34.4/32)

Doesn't know:

192.168.1.0/24
192.168.12.0/24
192.168.13.0/24



Default Gateway

Routing Packets: Default Gateway

- End hosts like PC1 and PC4 can send packets directly to destinations in their connected network.
→ PC1 is connected to 192.168.1.0/24, PC4 is connected to 192.168.4.0/24.

- To send packets to destinations outside of their local network, they must send the packets to their **default gateway**.

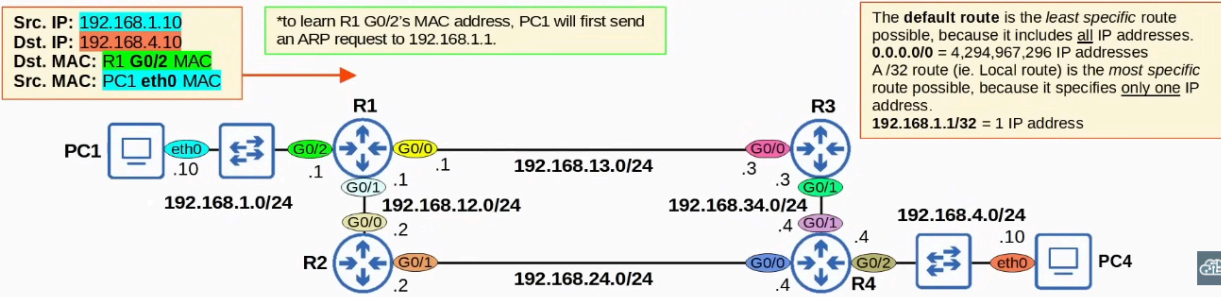
PC1 (Linux) Config:

```
iface eth0 inet static
address 192.168.1.10/24
gateway 192.168.1.1
```

PC4 (Linux) Config:

```
iface eth0 inet static
address 192.168.4.10/24
gateway 192.168.4.4
```

- The **default gateway** configuration is also called a **default route**.
→ It is a route to 0.0.0.0/0 = all netmask bits set to 0. Includes all addresses from 0.0.0.0 to 255.255.255.255.
- End hosts usually have no need for any more specific routes.
→ They just need to know: to send packets outside of my local network, I should send them to my default gateway.



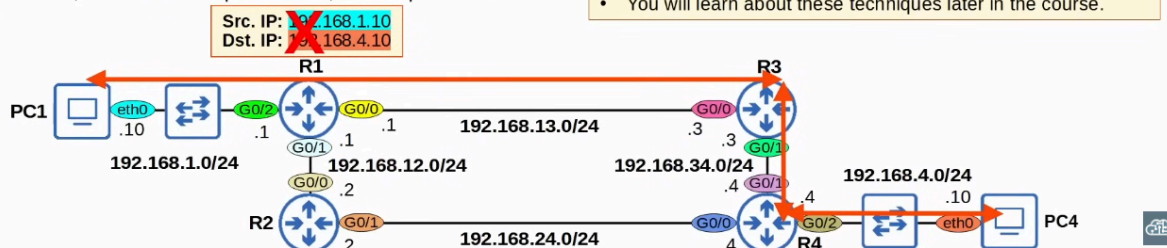
STATIC ROUTES:

Routing Packets: Static Routes

- When R1 receives the frame from PC1, it will de-encapsulate it (remove L2 header/trailer) and look at the inside packet.
- It will check the routing table for the most-specific matching route:
- R1 has no matching routes in its routing table.
→ It will drop the packet.
- To properly forward the packet, R1 needs a route to the destination network (192.168.4.0/24).
→ Routes are instructions: *To send a packet to destinations in network 192.168.4.0/24, forward the packet to next hop Y.*
- There are two possible path packets from PC1 to PC4 can take:
 - 1) PC1 → R1 → R3 → R4 → PC4
 - 2) PC1 → R1 → R2 → R4 → PC4
- In this video, we will use the path via R3, not the path via R2.

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, GigabitEthernet0/2
L 192.168.1.1/32 is directly connected, GigabitEthernet0/2
C 192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected, GigabitEthernet0/1
L 192.168.12.1/32 is directly connected, GigabitEthernet0/1
C 192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.13.0/24 is directly connected, GigabitEthernet0/0
L 192.168.13.1/32 is directly connected, GigabitEthernet0/0
```

- It is possible to configure the routers to:
→ *load-balance* between path 1) and 2)
→ Use path 1) as the main path and path 2) as a backup path
- You will learn about these techniques later in the course.



STATIC ROUTE CONFIGURATION:

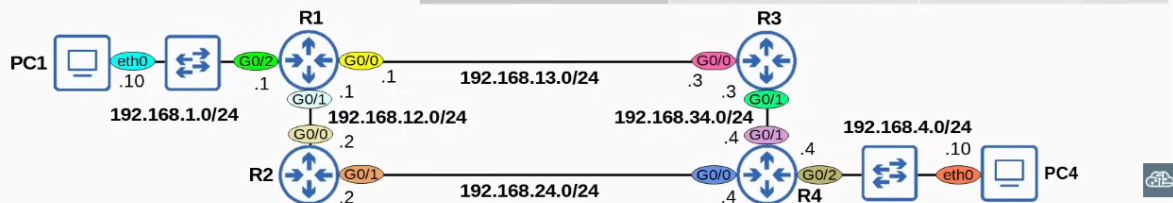
Static Route Configuration

- Each router in the path needs two routes: a route to 192.168.1.0/24 and a route to 192.168.4.0/24.
→ This ensures **two-way reachability** (PC1 can send packets to PC4, PC4 can send packets to PC1).
- R1 already has a **Connected route** to 192.168.1.0/24. R4 already has a **Connected route** to 192.168.4.0/24.
→ The other routes must be manually configured (using **Static routes**).

*routers don't need routes to all networks in the path to the destination.
→ R1 doesn't need a route to 192.168.34.0/24.
→ R4 doesn't need a route to 192.168.13.0/24.

- To allow PC1 and PC4 to communicate with each other over the network, let's configure these **Static routes** on R1, R3, and R4.

Router	Destination	Next-Hop
R1	192.168.1.0/24	Connected
	192.168.4.0/24	192.168.13.3
R3	192.168.1.0/24	192.168.13.1
	192.168.4.0/24	192.168.34.4
R4	192.168.1.0/24	192.168.34.3
	192.168.4.0/24	Connected



Static Route Configuration (R1)

R1(config)# ip route 192.168.4.0 255.255.255.0 192.168.13.3 R1(config)# ip route ip-address netmask next-hop

R1(config)# do show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
!some code output omitted

Gateway of last resort is not set

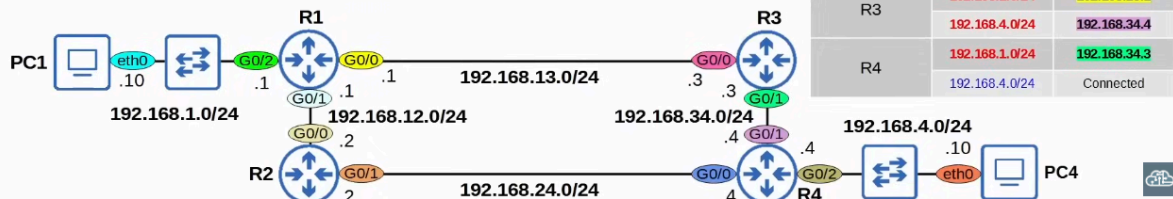
```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet0/2
S    192.168.4.0/24 [1/0] via 192.168.13.3
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, GigabitEthernet0/1
L    192.168.12.1/32 is directly connected, GigabitEthernet0/1
C    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/0
L    192.168.13.1/32 is directly connected, GigabitEthernet0/0

```

The [1/0] displayed in static routes means:
[Administrative Distance/Metric]
We will cover these concepts later in the course.

Router	Destination	Next-Hop
R1	192.168.1.0/24	Connected
	192.168.4.0/24	192.168.13.3
R3	192.168.1.0/24	192.168.13.1
	192.168.4.0/24	192.168.34.4
R4	192.168.1.0/24	192.168.34.3
	192.168.4.0/24	Connected



Static Route Configuration (R3)

```
R3(config)# ip route 192.168.1.0 255.255.255.0 192.168.13.1
R3(config)# ip route 192.168.4.0 255.255.255.0 192.168.34.4
```

```
R3(config)# ip route ip-address netmask next-hop
```

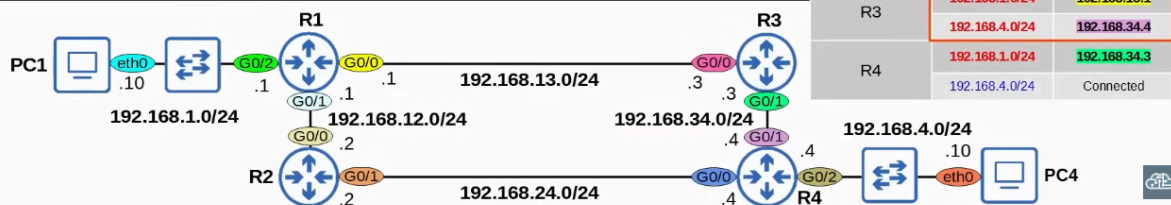
```
R3(config)# do show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
!some code output omitted
```

Gateway of last resort is not set

```
S    192.168.1.0/24 [1/0] via 192.168.13.1
S    192.168.4.0/24 [1/0] via 192.168.34.4
C    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/0
L    192.168.13.3/32 is directly connected, GigabitEthernet0/0
L    192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet0/1
L    192.168.34.3/32 is directly connected, GigabitEthernet0/1
```

Router	Destination	Next-Hop
R1	192.168.1.0/24	Connected
	192.168.4.0/24	192.168.13.3
R3	192.168.1.0/24	192.168.13.1
	192.168.4.0/24	192.168.34.4
R4	192.168.1.0/24	192.168.34.3
	192.168.4.0/24	Connected



Static Route Configuration (R4)

```
R4(config)# ip route 192.168.1.0 255.255.255.0 192.168.34.3
```

```
R4(config)# ip route ip-address netmask next-hop
```

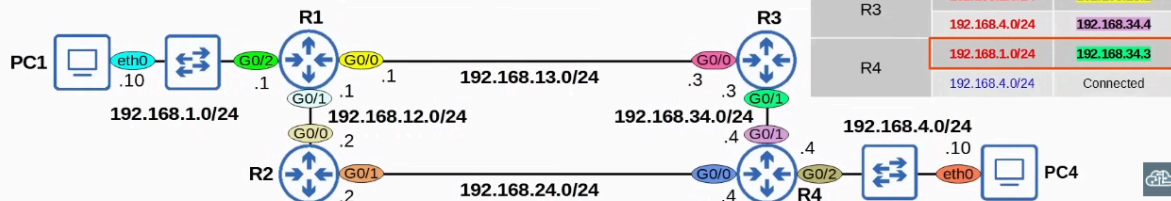
```
R4(config)# do show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
!some code output omitted
```

Gateway of last resort is not set

```
S    192.168.1.0/24 [1/0] via 192.168.34.3
C    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/2
L    192.168.4.4/32 is directly connected, GigabitEthernet0/2
C    192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.24.0/24 is directly connected, GigabitEthernet0/0
L    192.168.24.4/32 is directly connected, GigabitEthernet0/0
C    192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, GigabitEthernet0/1
L    192.168.34.4/32 is directly connected, GigabitEthernet0/1
```

Router	Destination	Next-Hop
R1	192.168.1.0/24	Connected
	192.168.4.0/24	192.168.13.3
R3	192.168.1.0/24	192.168.13.1
	192.168.4.0/24	192.168.34.4
R4	192.168.1.0/24	192.168.34.3
	192.168.4.0/24	Connected



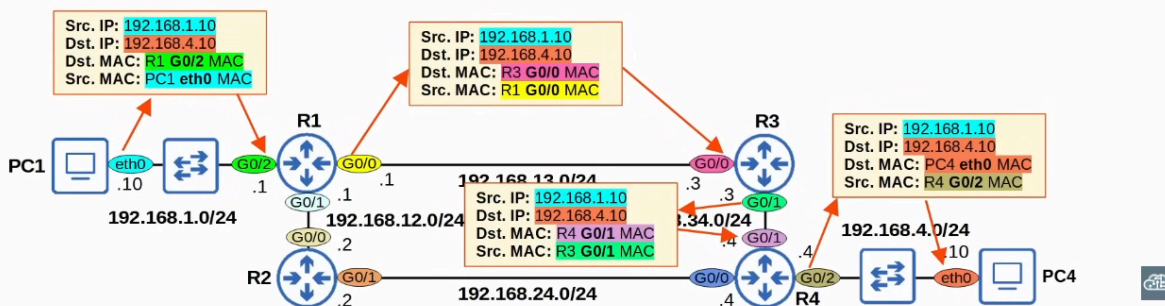
PC1 ↔ PC4

```
PC1:~$ ping 192.168.4.10
PING 192.168.4.10 (192.168.4.10): 56 data bytes
64 bytes from 192.168.4.10: seq=0 ttl=42 time=8.745 ms
64 bytes from 192.168.4.10: seq=1 ttl=42 time=4.423 ms
64 bytes from 192.168.4.10: seq=2 ttl=42 time=3.428 ms
64 bytes from 192.168.4.10: seq=3 ttl=42 time=3.544 ms
64 bytes from 192.168.4.10: seq=4 ttl=42 time=3.520 ms
^C
--- 192.168.4.10 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 3.428/4.732/8.745 ms
```

If the ping is successful, that means there is two-way reachability.
PC1 can reach PC4, and PC4 can reach PC1.

Packet traveling from PC1 to PC4:

*we will examine this step-by-step in the "Life of a Packet" video



STATIC ROUTE CONFIGURATION with *exit-interface*

Static Route Configuration with exit-interface

```
R2(config)# ip route 192.168.1.0 255.255.255.0 g0/0
R2(config)# ip route 192.168.4.0 255.255.255.0 g0/1 192.168.24.4
```

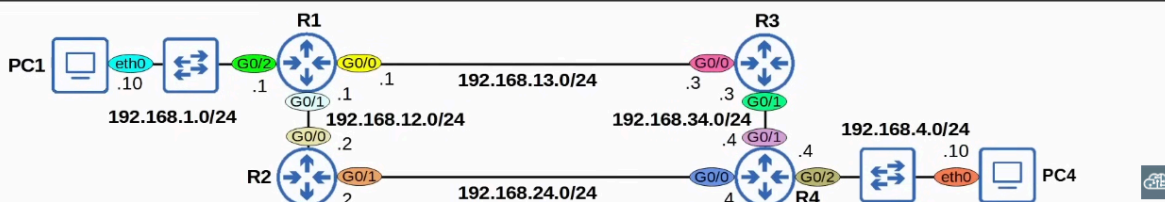
```
R2(config)# do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
!some code output omitted
```

Gateway of last resort is not set

```
S 192.168.1.0/24 is directly connected, GigabitEthernet0/0
S 192.168.4.0/24 [1/0] via 192.168.24.4, GigabitEthernet0/1
C 192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.12.0/24 is directly connected, GigabitEthernet0/0
L 192.168.12.2/32 is directly connected, GigabitEthernet0/0
C 192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.24.0/24 is directly connected, GigabitEthernet0/1
L 192.168.24.2/32 is directly connected, GigabitEthernet0/1
```

```
R2(config)# ip route ip-address netmask exit-interface
R2(config)# ip route ip-address netmask exit-interface next-hop
```

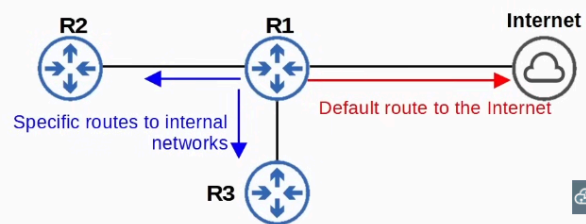
- Static routes in which you specify only the *exit-interface* rely on a feature called **Proxy ARP** to function.
- This is usually not a problem, but generally you can stick to *next-hop* or *exit-interface next-hop*.
- Neither is 'better' than the other; use which you prefer.



DEFAULT ROUTE

Default Route

- A **default route** is a route to 0.0.0.0/0
→ 0.0.0.0/0 is the *least specific* route possible; it includes every possible destination IP address.
- If the router doesn't have any more specific routes that match a packet's destination IP address, the router will forward the packet using the **default route**.
- A default route is often used to direct traffic to the Internet.
→ More specific routes are used for destinations in the internal corporate network.
→ Traffic to destinations outside of the internal network is sent to the Internet.

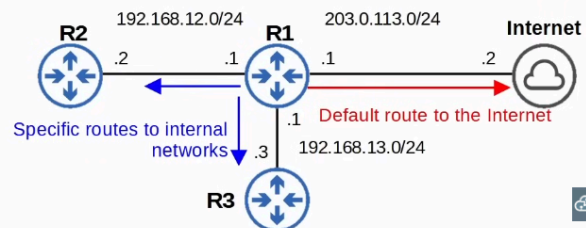


Default Route

```
R1# show ip route
!codes omitted
Gateway of last resort is not set
```

Destination	Next Hop	Interface
10.0.0.0/8	192.168.12.2	GigabitEthernet0/1
172.16.0.0/16	192.168.13.3	GigabitEthernet0/1
192.168.12.0/24	192.168.12.1	GigabitEthernet0/1
192.168.12.1/32	192.168.12.1	GigabitEthernet0/1
192.168.13.0/24	192.168.13.1	GigabitEthernet0/0
192.168.13.1/32	192.168.13.1	GigabitEthernet0/0
203.0.113.0/24	203.0.113.1	GigabitEthernet0/2
203.0.113.1/32	203.0.113.1	GigabitEthernet0/2

No default route has been configured yet.




```
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)# do show ip route
!most codes omitted
  ia - IS-IS inter area, * - candidate default, U - per-user static route
!most codes omitted
Gateway of last resort is 203.0.113.2 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [1/0] via 203.0.113.2
S 10.0.0.0/8 [1/0] via 192.168.12.2
S 172.16.0.0/16 [1/0] via 192.168.13.3
C 192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
L 192.168.12.0/24 is directly connected, GigabitEthernet0/1
L 192.168.12.1/32 is directly connected, GigabitEthernet0/1
C 192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
L 192.168.13.0/24 is directly connected, GigabitEthernet0/0
L 192.168.13.1/32 is directly connected, GigabitEthernet0/0
C 203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
L 203.0.113.0/24 is directly connected, GigabitEthernet0/2
L 203.0.113.1/32 is directly connected, GigabitEthernet0/2
```

